

Low-redundancy motor imagery EEG decoding based on dynamic attention and feature reconstruction

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ABSTRACT

Electroencephalography (EEG) motor imagery (MI) decoding is a critical component in the development of motor imagery brain-computer interfaces (MI-BCI). The decoding performance reflects the ability to accurately identify and classify brain activity from EEG signals, which is essential during motor imagery tasks. A major challenge lies in accurately recognizing human intent from EEG-based MI (MI-EEG) brain signal that possess low signal-to-noise ratios and instability. The effectiveness of MI classification directly depends on decoding performance, impacting the practicality and reliability of BCI systems. To address this challenge, we investigate a dynamic combinable attention-based spatial and channel reconstruction convolutional network (DCA-SCRCNet) that leverages redundant features in weak MI-EEG signals to optimize spatial distribution features and inter-channel interactions relationships. This optimization is achieved through weight-based feature reconstruction in the spatial and channel reconstruction convolution (SCRC) module. Importantly, dynamic combinable attention (DCA) enables DCA-SCRCNet to adapt to differences in the non-stationary MI-EEG signals individual characteristics by weighting input features and integrating local and global information. The processed features are then fed into the temporal convolutional network (TCN) for further extraction of long-term dependencies and finer timing characteristics. On the BCI Competition IV-2a dataset (BCI-2a), subject-independent and subject-dependent models yielded accuracies of 70.7 % and 90.5 %, respectively, demonstrating its efficient spatio-temporal feature extraction capability while reducing computational redundancy. On the PhysioNet dataset, the proposed algorithm achieved multi-subject validation accuracies of 92.7 % and 76.2 % for two and four-class classification tasks, respectively, surpassing current mainstream methods.

1. Introduction

Brain-Computer Interface (BCI) technology, serving as a bridge between the brain and external devices, has made significant progress in fields such as medicine, rehabilitation, and intelligent control systems [1]. Specifically, in motor imagery (MI) tasks, decoding EEG signals related to MI enables researchers to provide effective assistive tools for stroke patients, individuals with disabilities, and others requiring alternative control systems. MI EEG signals, generated through activation of motor-related brain regions during imagined movements, exhibit distinct spatio-temporal patterns characterized by event-related

desynchronization/synchronization (ERD/ERS) of μ (8–12 Hz) and β (13–30 Hz) rhythms [2]. However, the inherent low signal-to-noise ratio (SNR), non-stationarity, and inter-subject variability of MI-EEG signals pose significant challenges for decoding in MI tasks.

To enable robust and generalizable decoding, it is essential to address three fundamental challenges that are often underexplored in existing work. First, temporal misalignment arises from trial-specific variability in rhythm dominance (μ vs. β), onset latency, and duration, leading to inconsistency in the temporal location of informative neural patterns. Fixed-size receptive fields or static attention mechanisms struggle to align features across trials, especially under non-stationary

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dynamics. Second, feature redundancy exists in both the input signals and the deep models. High-dimensional multi-channel EEG data often contain spatially overlapping and spectrally correlated components [3]. Deep convolutional networks may further amplify redundancy through stacked filters that capture repetitive or irrelevant features, leading to inefficiency and degraded performance. Third, inter-subject variability, rooted in anatomical, cognitive, and physiological differences,

results in distributional shifts between training and test subjects, hindering model generalization and necessitating expensive subject-specific calibration.

Traditional machine learning (ML) pipelines typically follow a three-stage process—preprocessing, feature extraction, and classification—but often rely on handcrafted features and fixed filter banks that fail to adapt to individual-specific dynamics. Due to the low SNR and susceptibility to artifacts, methods like Independent Component Analysis [4] can separate artifacts, but their success depends on expert experience, and incorrect artifact removal risks losing critical MI signals. ML approaches use fixed feature extraction strategies and classifiers, limiting adaptability to individual variability, ultimately affecting classification performance. For instance, Filter Bank Common Spatial Pattern and its variants [5] decompose EEG signals into multiple frequency bands and apply spatial filters, but they rely on manually defined frequency bands and filters, showing limited adaptability to individual differences. Moreover, MI-EEG signals have low temporal and frequency resolution [6] with complex spatio-temporal patterns, stemming from interactions between spatial and temporal features and dynamic changes across frequency bands. Spatial features, recorded from scalp electrodes, represent aggregated brain region activity, with spatial complexity reflecting interactions among brain regions. Manual feature extraction struggles to capture the rich, multidimensional spatio-temporal information in MI signals.

In recent years, deep learning (DL) has been widely adopted in signal decoding tasks, offering the ability to automatically learn spatio-temporal features from MI-EEG data and achieve end-to-end MI task classification [1]. Researchers have proposed several CNN variants, incorporating residual networks [7], 3D convolutions [7], multi-layer convolutions [8], and attention mechanisms [9], to improve performance.

However, CNNs focus primarily on local spatial features and often struggle to effectively capture the complex temporal dynamics inherent in MI-EEG signals. Pan et al. [10] introduced a shallow end-to-end convolutional network that was trained from scratch for pixel-wise saliency prediction. Their research showed that shallow architectures can capture spatially localized and interpretable features while maintaining low computational complexity. This is especially advantageous in scenarios with limited training data or noisy signals for MI-EEG decoding. Schirrmester et al. [10] promoted the application of shallow and deep convolutional architectures in EEG decoding. The fundamental concept of using shallow convolutional networks to learn compact and localizable patterns has been extensively explored in broader vision tasks. In addition, Schirrmester et al. [11] introduced two architectures, Deep ConvNet and Shallow ConvNet, to capture complex spatio-temporal patterns in MI-EEG signals. Deep ConvNet, with its deeper architecture, can extract complex spatial features but often over-extracts irrelevant ones due to the limited EEG data sample size, increasing computational overhead. Group-wise convolution (GWC), which was first introduced in AlexNet [12], can be regarded as a sparse convolution connection method that each output channel is connected to only a certain group of input channels, a variant of the standard convolution, are proposed to replace the existing expensive convolutional operation.

Despite recent progress, existing deep learning methods still struggle with pervasive feature redundancy, which arises from multiple sources and hinders effective MI-EEG decoding [12]. At the signal level, EEG recordings suffer from volume conduction, causing strong spatial correlations across adjacent electrodes and overlapping frequency-band responses (e.g., co-occurring μ and β ERD), especially in sensorimotor

areas [13]. Existing methods such as [14] [15] only handle single-dimensional redundancy. OctConv [14] targets spatial redundancy but cannot process cross-band channel redundancy. GhostNet [15] addresses channel redundancy while neglecting spatial locality. At the model level, stacked convolutional layers tend to repeatedly extract similar spatio-temporal patterns across depths, while static spatial filters and attention weights often assign importance to redundant or physiologically irrelevant regions [16]. Hybrid architectures (e.g., CNN-RNN) may also reprocess already learned features, leading to redundant representations across modules. Traditional dimensionality reduction (PCA) [17] loses nonlinear ERD/ERS features through linear compression. At the data level, limited MI-EEG samples restrict feature diversity during training, pushing the model toward overfitting on easy but sub-optimal patterns and converging to highly similar filters across layers [18]. However, these compression-optimized networks have mainly been used for image classification and object detection, and further research is needed to verify their effectiveness in low-SNR MI-EEG signal and spatio-temporal feature extraction.

While CNNs excel at capturing spatial features in EEG signals, they struggle with modeling temporal dependencies across time windows and frequency bands, which is crucial for MI tasks where temporal variability demands robust temporal modeling. To address this, researchers have proposed models based on Temporal Convolutional Network (TCN) [19]. TCNs use dilated and causal convolutions to exponentially expand the receptive field while linearly increasing the number of parameters, avoiding issues such as vanishing or exploding gradients. Compared to traditional recurrent neural networks (RNNs) like LSTM and GRU [19], TCNs demonstrate superior performance in sequence tasks due to their parallel computation capabilities. Ingolfsson et al. [13] proposed the EEG-TCNet architecture, which leverages TCNs to capture long-term dependencies in EEG signals while enhancing spatial feature extraction with EEGNet, addressing MI-EEG nonstationarity. TCNs are effective in capturing long-term dynamics by expanding the receptive field, making them suitable for handling temporal changes in MI tasks. However, TCNs mainly focus on temporal modeling and lack the capability to capture spatial information, limiting their ability to fully integrate spatio-temporal features. Multi-Head Attention (MHA) is a key component of the Transformer model [15], captures global spatio-temporal dependencies in EEG signals. However, it struggles to capture local features, which are crucial for MI-EEG classification, potentially leading to information loss. To address the spatio-temporal dynamics of nonstationary MI-EEG signals, some researchers have employed CNNs to capture local spatial information, though this often increases computational costs. Lawhern et al. [12] proposed EEGNet, which combines depthwise separable convolutions (DWC) and point-wise convolutions (PWC) to apply convolutional filters independently to each EEG channel, extracting

local spatial features and integrate features across channels, effectively capturing spatial relationships between different brain regions. For instance, Amin et al. [21] combined attention layers with Inception-CNN and LSTM. Compared to traditional CNNs, Inception-CNN uses convolution kernels of varying sizes to capture multi-scale EEG features, but this increases model complexity and computational cost. Deng et al. [22] effectively captured the multi-level dynamic information in EEG signals by integrating a spatiotemporal attention mechanism, multi-level wavelet convolutional blocks, and Graph Convolutional Networks (GCNs). Additionally, Altafheri et al. [23] combined the temporal dependency modeling ability of TCNs, the temporal and local spatial feature extraction capability of CNNs, and the global feature fusion ability of attention mechanisms, significantly improving classification performance in MI tasks. However, despite performance improvements, multi-layer convolutions may introduce redundant features in MI tasks. The Multi-Head Self-Attention (MSA) mechanism in [23] effectively captures global information, yet it may overemphasize global features and neglect local details, potentially missing subtle individual differences. Additionally, these mechanisms

have high computational complexity, affecting real-time processing capabilities of the model.

Despite significant progress in MI-EEG decoding, existing models often suffer from several limitations. Most methods focus on extracting spatial or spectral features independently, without explicitly addressing cross-dimensional redundancies that arise from overlapping or correlated patterns across spatial and frequency domains. Moreover, conventional attention mechanisms typically apply fixed parameterizations across all trials, ignoring the inherent variability in temporal dynamics among subjects and sessions. Many architectures also rely on deep or multi-stage pipelines, resulting in increased redundancy, limited generalization, and high computational costs. To overcome these issues, we propose a unified framework that incorporates redundancy-aware feature refinement, trial-adaptive attention, and efficient architectural design.

Rather than simply applying ML/DL, we develop a new class of neuro-adaptive architectures to address the aforementioned issues, dynamic combinable attention-based spatial and channel reconstruction convolutional network (DCA-SCRCNet), which aims to fully extract spatio-temporal and individual-specific features from MI-EEG signal and improve decoding accuracy without increasing computational cost or parameters. The main contributions of our study are summarized below:

- (1) We employ gradient magnitudes obtained from non-linear group normalization layers to assess spatial importance. This assessment guides the adaptive partitioning of EEG channels into informative and redundant subsets through a learnable gating matrix. Discriminative channels are refined using group-wise and point-wise convolutions, while redundant channels are softly merged to preserve auxiliary cues. This cross-dimensional reconstruction allows the network to suppress irrelevant spatial patterns and enhance task-relevant activations.
- (2) We propose a novel attention composition strategy that reweights temporal heads according to the trial-specific rhythm power. This allows for the adaptive adjustment of receptive fields across temporal scales. For example, during the early onset of motor imagery (0.5–2 s), fine-resolution heads are emphasized, while during the steady-state periods, coarse-resolution integration prevails. The composition weights are modulated in accordance with transient event-related desynchronization (ERD) or event-related synchronization (ERS) patterns. This enables the model to flexibly capture both short bursts and sustained rhythms without making fixed assumptions.
- (3) By integrating channel-selective gating and low-rank dynamic projections into a single trainable network, DCA-SCRCNet performs end-to-end feature pruning and adaptive fusion in both spatial and temporal domains. Its lightweight design leverages depth-efficient convolutions and low-rank attention to minimize

parameter count and inference cost, while still achieving strong generalization across subjects, as validated by our experiments on BCI Competition IV-2a and PhysioNet datasets, where DCA-SCRCNet achieves 70.7 %/90.5 % (subject-independent/dependent) and 92.7 %/76.2 % (2-class/4-class) classification accuracies, significantly outperforming existing approaches.

2. Methods

To address the challenges of non-stationarity, feature redundancy, and inter-

subject variability in MI-EEG decoding, we propose a domain-specific, input-adaptive neural framework named DCA-SCRCNet, as illustrated in Fig. 1. The model is composed of three core modules that work collaboratively and progressively to extract high-level representations: a spatial-channel reconstruction convolution (SCRC) module, which simulates selective activation in the sensorimotor cortex by enhancing motor-relevant electrodes and suppressing redundancy across spatial and channel dimensions; a dynamic combinable attention (DCA) module, which adaptively adjusts attention heads per trial to accommodate temporal variations in oscillatory rhythms (e.g., μ/β ; and a temporal convolutional network (TCN), which captures sequential dependencies and complements the attention mechanism. These components are unified into a lightweight, end-to-end trainable architecture that achieves both accurate and generalizable MI-EEG decoding with competitive efficiency. In the following sections, we detail the key components of the proposed model.

2.1. Data augmentation

The raw MI-EEG signals $\mathbf{X}_a \in \mathbb{R}^{C \times \bar{T}}$, represented as a matrix with C channels (EEG electrodes) and $\bar{T}$ time points. The goal is to map the input MI-EEG signals to their corresponding MI classes. The dataset contains $\tilde{n}$ labeled MI trials $S = \{\mathbf{X}_a, y_a\}_{a=1}^{\tilde{n}}$, where $y_a \in \{1, \dots, N\}$ denotes the labels, and N represents the number of MI classes.

Given the MI-EEG signals with C channels, each channel containing $\bar{T}$ time points, data augmentation techniques are applied firstly to enhance the ability to capture the spatio-temporal dynamics of MI-EEG signals (Sees details in Section III.C). Specifically, we apply time shifting, randomly shifting the signals along the time axis within a range of ± 10 time points (max_shift), leveraging the strong temporal dependency of MI-EEG signals.

Inter-subject variability in the amplitude of MI-EEG signals can negatively impact performance of the model. To mitigate this, a scaling operation is applied to adjust the signal amplitude within 90–110 % of the original amplitude ($scale = 0.1$). MI-EEG signals are susceptible to external interference, such as noise from environmental factors or the

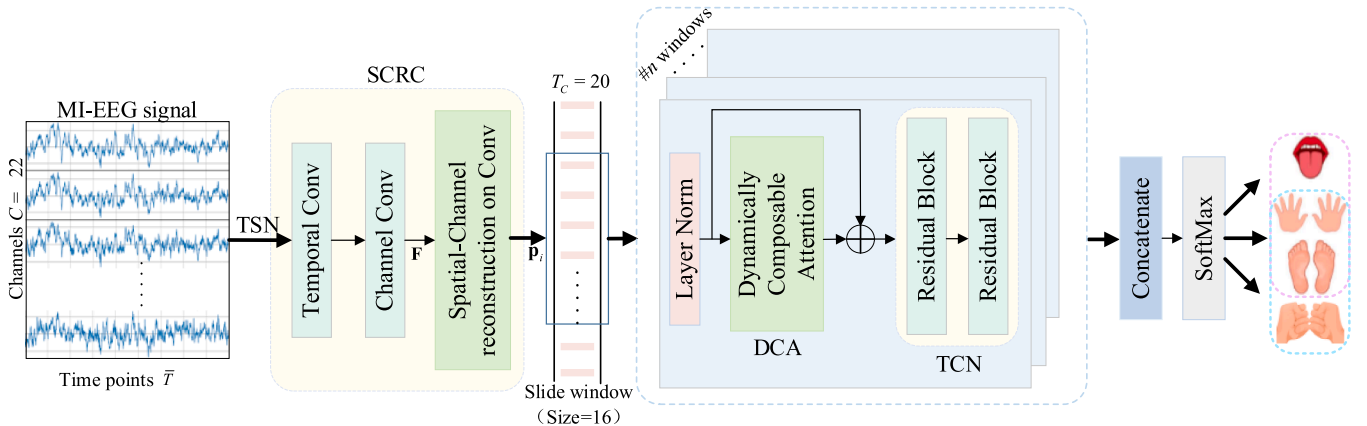


Fig. 1. Components of the DCA-SCRCNet.

measurement process. To simulate real-world conditions and improve robustness of the model to such noise, we add gaussian noise, with a mean of 0 and a standard deviation controlled by a *noise level* parameter. TSN helps to simulate diverse and realistic scenarios, which improve model generalization and robustness.

The augmented dataset $\tilde{S} = \{(\tilde{X}_a^b, y_a)\}_{a=1, b=1}^{\tilde{n}, \tilde{k}}$ includes $\tilde{k}$ augmented versions of each original trial X_a , where $\tilde{X}_a^b$ denotes the augmented version of trial X_a . After augmentation, the total number of trials in the dataset increases to $\tilde{n} \times (\tilde{k}+1)$. Each augmented trial maintains the original structure of C channels and $\bar{T}$ time points, but the signal contents are altered by the augmentation process.

2.2. Spatial and channel reconstruction convolution (SCRC)

In the proposed architecture, feature extraction and temporal modeling are jointly designed to align with the underlying sensorimotor rhythms. In the Feature Extraction Stage, the cross-reconstruction mechanism targets both signal-level redundancy (e.g., spatially overlapping activations, correlated frequency bands) and model-level convolutional redundancy by reconstructing and fusing features across electrode and frequency domains. We utilize gradient magnitudes derived from non-linear Group Normalization layers as spatial importance cues. This enables adaptive partitioning of EEG channels into informative and redundant groups using a learnable spatial gating matrix. Task-relevant sensorimotor regions are preserved and enhanced via threshold-guided reconstruction and cross-group fusion. A predefined ratio α is used to split channels into discriminative and redundant subsets. Discriminative channels are refined via group-wise and point-wise Convolutions (GWC, PWC), while redundant channels are preserved to retain auxiliary cues. These features are then softly fused using attention weighting to maintain diversity while reducing noise.

The SCRC module consists of temporal convolution, channel convolution, and spatial-channel reconstruction convolution, as illustrated in Fig. 2 (using the BCI-2a dataset as an example). Temporal convolution and channel convolution use the same structure and parameters as in [23]. Temporal convolution captures temporal dynamics along the time axis of length $\bar{T}$ and sampling rate F_s , producing E_1 sequential feature maps of size $\bar{T} \times C$. Channel convolution has D filters, integrating global features across the temporal feature maps while focusing on inter-electrode channel correlations, outputting E_2 feature

maps of size $\bar{T} \times 1$. To mitigate potential information loss from compression and provide complementary spatial features for motor imagery decoding, we implement a divide-and-conquer approach, including spatial-channel reconstruction convolution branch and spatial convolution branch. One branch performs redundant removal and feature reduction through channel and spatial reconstruction convolution while the other branch applies spatial convolution to supplement inter-electrode spatial distribution. The spatial convolution does not perform redundancy reduction, thus it maintaining integrity of the original spatial distribution. The outputs of both branches are combined element-wise, with the redundancy-reduction branch contributing task-relevant features, and the spatial convolution branch retains spatial distribution integrity, offering complementary advantage. The final output consists of T_C temporal sequences, each with dimensionality $d = E_2$.

In spatial-channel reconstruction convolution branch, a 1×1 convolution is used first to restore the number of channels in a feature map F , enabling subsequent operations to emphasize critical features. The spatial information richness of feature maps is then evaluated using the trainable parameter γ of the group normalization (GN) layer. The weights W_γ obtained from GN are used to “separate” the feature map into informative and redundant components. The operation mathematically defined as:

$$F_{out} = GN(F) = \gamma \frac{F - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta \quad (1)$$

where μ and σ are the mean and standard deviation of F , respectively, ϵ is a small constant added for numerical stability, γ and β are trainable affine transformation parameters. The γ values for all channels are normalized to obtain the weight for each channel W_γ :

$$W_\gamma = \{w_i\} = \frac{\gamma_i}{\sum_{j=1}^C \gamma_j}, i, j = 1, 2, \dots, C \quad (2)$$

where W_γ are mapped to the range of (0,1) using a Sigmoid function, followed by a threshold-based separation operation:

$$W = \text{Gate}(\text{Sigmoid}(W_\gamma(\text{GN}(F)))) \quad (3)$$

The redundant features are separated as follows

$$F_1^o = F_1 \otimes F \quad (4)$$

$$F_2^o = F_2 \otimes F \quad (5)$$

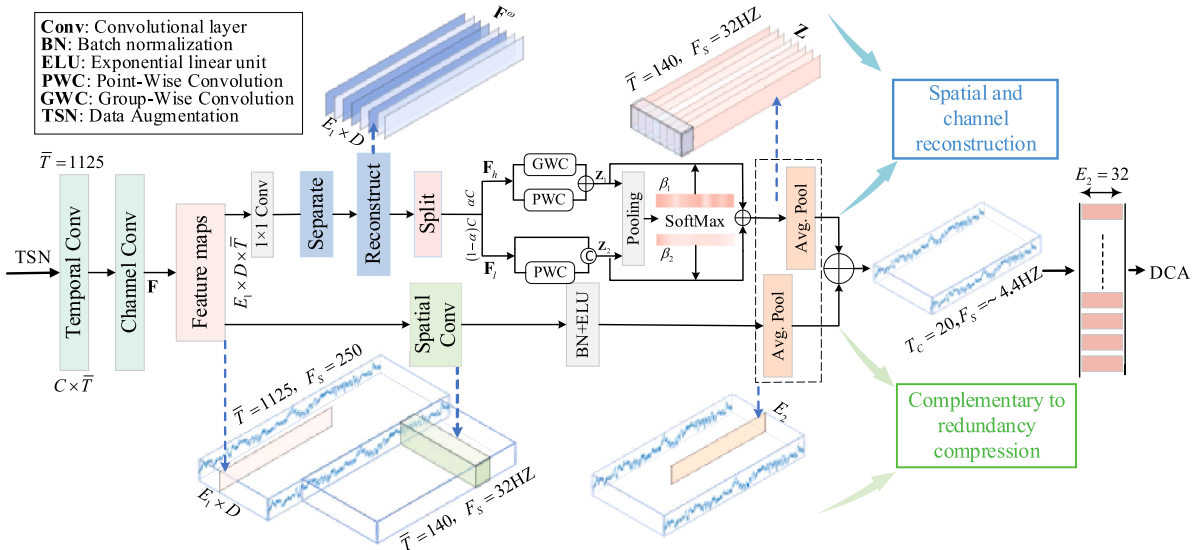


Fig. 2. The SCRC block eliminates redundant features in the spatial and channel dimensions through spatial and channel reconstruction convolutions, and complements the spatio-temporal encoding with spatial convolutions.

where $\otimes$ denotes element-wise multiplication, ensuring redundant features $\mathbf{F}_1^o$ like noise or irrelevant data are zeroed out while retaining only the most informative features $\mathbf{F}_2^o$ from distinct brain regions. Instead of directly summing these components, we apply “reconstruction” to facilitate information flow between $\mathbf{F}_1^o$ and $\mathbf{F}_2^o$:

$$\mathbf{F}^{o1} = \mathbf{F}_{11}^o \oplus \mathbf{F}_{22}^o \quad (6)$$

$$\mathbf{F}^{o2} = \mathbf{F}_{21}^o \oplus \mathbf{F}_{12}^o \quad (7)$$

$$\mathbf{F}^o = \text{Concat}(\mathbf{F}^{o1}, \mathbf{F}^{o2}) \quad (8)$$

where $\oplus$ is element-wise summation, the reconstructed features are concatenated, producing a more comprehensive and representative feature map $\mathbf{F}^o$.

To address redundant channel features, $\mathbf{F}^o$ is “split” into two components: $\mathbf{F}_h$ for extracting information-rich features and $\mathbf{F}_l$ for processing low-level redundant features. We use GWC to divide channels in $\mathbf{F}_h$ into groups randomly, and convolution is performed independently within each group. This preserves local feature relationships while significantly reducing the number of parameters. However, GWC inherently limits inter-group channel interactions, as it isolates information flow across groups. To compensate for this limitation, PWC is employed. PWC uses 1×1 convolution, which maintains spatial resolution while performing a weighted summation across all channels. The final merged feature map $\mathbf{Z}_1$ is defined as:

$$\mathbf{Z}_1 = \mathbf{M}^G \mathbf{F}_h + \mathbf{M}^{P1} \mathbf{F}_l \quad (9)$$

where, $\mathbf{M}^G$ and $\mathbf{M}^{P1}$ are the learnable weights of GWC and PWC, respectively. To further refine the representation, a simple 1×1 convolution is applied to $\mathbf{F}_l$, allowing the reuse of $\mathbf{F}_l$ without additional cost. The transformed features are then concatenated, forming the low-level stage output $\mathbf{Z}_2$:

$$\mathbf{Z}_2 = \mathbf{M}^{P2} \mathbf{F}_l \cup \mathbf{F}_l \quad (10)$$

where $\mathbf{M}^{P2}$ is a learnable matrix.

Following this transformation, an attention-base fusion mechanism [18] is applied to adaptively merge the processed features. Specifically, “global average pooling” is used to generate channel-level feature descriptors $\mathbf{S}_m \in \mathbb{R}^{C \times 1 \times 1}$:

$$\mathbf{S}_m = \text{Pooling}(\mathbf{Z}_m) = \frac{1}{\bar{T} \times C} \sum_{e_1=1}^{\bar{T}} \sum_{e_2=1}^{\bar{C}} \mathbf{Z}_m(e_1, e_2), m = 1, 2 \quad (11)$$

$$\mathbf{Z} = \mathbf{Z}_1 \beta_1 + \mathbf{Z}_2 \beta_2 \quad (12)$$

where $\bar{T}$ is the temporal dimension of the feature map and C stands for channels. Soft attention weights β_1, β_2 are used to guide the fusion of the two feature sets, automatically selecting which features contribute most to decoding MI-EEG signals, and automatically optimizing the output $\mathbf{Z}$.

The spatial convolution branch employs spatial convolution with $E_2 = E_1 \times D$ filters sized $(1, K_{C2})$ to supplement the redundancy reduction branch. Spatial convolution uses the same structure as in [23]. Average (Avg) pooling further reduces the sampling rate while maintaining important spatial information, ensuring that the processed signals are more interpretable by the model. Batch normalization (BN) [24] is applied to convolution layers to accelerate training, while the exponential linear line (ELU) activation function introduces non-linearity.

By aligning outputs of the two branches through dynamic pooling, the reconstructed features $\mathbf{Z}$ integrate spatial, channel, and temporal information. This design ensures that the final output $\mathbf{p}_i \in \mathbb{R}^{T_C \times d}$, represented as a sequence of temporal vectors of length T_C , each of length $d = E_2$, balances feature efficiency and discriminative power.

2.3. Dynamically combination multi-head attention (DCA)

In the Temporal Modeling Stage, dynamic attention composition modulates temporal window sensitivity and head importance, allowing the model to adapt to physiological rhythms (μ/β) and trial-specific dynamics. Unlike fixed attention

models, this configuration flexibly adapts to both within-subject and cross-subject variability. In this work, we apply Dynamic Combination Attention (DCA) to focus on local features that are helpful in MI tasks while simultaneously capturing global relationships across different time windows. The model dynamically reweights attention heads per trial, enabling automatic adjustment of temporal receptive fields. For instance, fine-grained heads are emphasized during MI onset (0.5–2 s), while coarse integration is preferred for steady-state periods. Composition weights are modulated based on transient power changes (e.g., ERD in μ/β bands). This ensures early evoked and late sustained components are captured by specialized heads and integrated into a unified representation. During training, the attention composition mechanism learns to adjust based on subject-specific input patterns, enhancing generalization across individuals without explicit recalibration.

In order to make the follow-up attention mechanism better focus on short-term temporal variations and local patterns related to the MI task, we employ a sliding window (SW) [23] to partition the output of SCRC ($\mathbf{p}_i$) into n small segments. Due to the distributional differences of features within different time windows, a Layer Normalization (Norm) is employed here to ensure stable input data.

In DCA, the attention vector $\mathbf{A}_{ij} = \mathbf{A}[:, i, j] \in \mathbb{R}^H$ represents the attention between a pair of query vectors $\mathbf{Q}_i \in D_m$ and key vectors $\mathbf{K}_j \in D_m$. Each head attention scores or weight matrices are dynamically composed and transformed into a new vector $\mathbf{A}'_{ij}$ with learnable parameters θ :

$$\mathbf{A}'_{ij} = \text{Compose}(\mathbf{A}_{ij}, \mathbf{Q}_i, \mathbf{K}_j; \theta) \quad (13)$$

where i and j denote the index positions of the query and key in the sequence and the trainable parameter $\theta = \{\mathbf{W}_b, \mathbf{W}_{q1}, \mathbf{W}_{q2}, \mathbf{W}_{qg}, \mathbf{W}_{k1}, \mathbf{W}_{k2}, \mathbf{W}_{kg}\}$.

This dynamic composition of attention scores (as shown in Fig. 3) helps the model learn diverse representations for different individuals, which is especially important for MI-EEG signals that exhibit high inter-subject variability. The attention vectors $\mathbf{A}_{ij}$ are split into five branches, each focusing on different aspects of the input signal. First, a static base weight matrix $\mathbf{W}_b$ is projected independently of $\mathbf{Q}_i$ or $\mathbf{K}_j$. To optimize it, a low-rank decomposition is applied to reduce the size of weight matrix and eliminate redundant parameters, ensuring a stable and efficient reference for the other branches. Other branches introduce dynamic projections like $\mathbf{w}_{q1} \in \mathbb{R}^{H \times R}$ to share attention weights across heads and gating mechanisms weight $\mathbf{W}_{qg}$ to control the degree of memory retention and feature selection in each head. The row-wise and column-wise components of $\mathbf{W}_q$ and

$\mathbf{W}_k$ are decomposed into lower-dimensional parts, thus reducing computational complexity. The row-wise decomposition focuses on the query matrix $\mathbf{Q}$, while the column-wise decomposition focuses on the key matrix $\mathbf{K}$, enabling efficient computation while preserving relevant feature. Through the row-plus-column decomposition, both the row-wise and column-wise components are further decomposed into low-rank and diagonal matrices, reducing the dimensionality of the projections from H to R . This reduces the overall complexity and enhances the efficiency without compromising its ability to capture relevant features in the attention space. The dynamic weights $\mathbf{w}_{q1}$ and $\mathbf{w}_{q2}$ are processed in chunks as follows:

$$\mathbf{w}_{q1}, \mathbf{w}_{q2} = \text{Chunk}(\text{GELU}(\mathbf{Q}_i \mathbf{W}_{q1}) \mathbf{W}_{q2}, \text{dim} = 1) \quad (14)$$

where $\mathbf{W}_{q1} \in \mathbb{R}^{D_m \times I}$ and $\mathbf{W}_{q2} \in \mathbb{R}^{I \times I}$ ($I=2HR$) are learnable weight matrices during model training, and a non-linear GELU activation

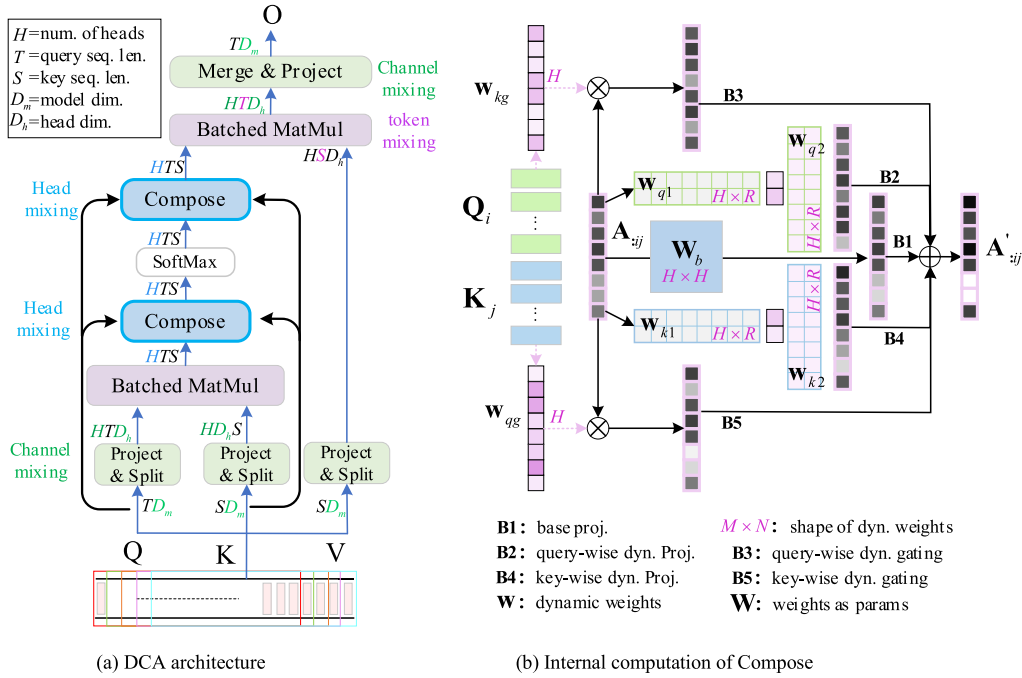


Fig. 3. Illustration of DCA. (a) Each linear projection's input and output are denoted by their dims and the projected (i.e. mixed) dims are colored. (b) Attention vector A_{ij} can be either attention scores or weights. In this paper, we use seq. len stands for sequence length, dyn. Proj. for dynamic project, dim. for dimension and num. for number.

function is applied to enhance the representation of the transformed query vector, adding non-linearity to the model. The dynamic gating weights computed as follows:

$$\mathbf{w}_{qg} = \tanh(\mathbf{Q}_i \mathbf{W}_{qg}) \quad (15)$$

where $\mathbf{W}_{qg} \in R^{D_m \times H}$ is a linear transformation matrix of $\mathbf{Q}_i$, and the hyperbolic tangent activation ($\tanh$) constrains the gate weights between $(-1, 1)$. Using low-rank plus diagonal decomposition, the gating mechanism can operate efficiently in

a smaller space, enhancing memory and feature selection while maintaining computational efficiency. By dynamically adjusting the gating weights, the model selectively suppresses redundant features and retains key information necessary for decoding MI-EEG signals, which often contain noise and irrelevant data. For the key vectors $\mathbf{K}_j$, a similar process is applied symmetrically, producing $\mathbf{w}_{k1}$, $\mathbf{w}_{k2}$ and $\mathbf{w}_{kg}$, and their respective transformations. The final output attention vector is obtained by summing the results of all five branches:

$$\begin{aligned} \mathbf{A}'_{ij} = & \mathbf{A}_{ij} \mathbf{W}_b + \mathbf{A}_{ij} \mathbf{w}_{q1} \mathbf{w}_{q2} + \mathbf{A}_{ij} \otimes \mathbf{w}_{qg} \\ & + \mathbf{A}_{ij} \mathbf{w}_{k1} \mathbf{w}_{k2} + \mathbf{A}_{ij} \otimes \mathbf{w}_{kg} \end{aligned} \quad (16)$$

where $\mathbf{W}_b, \mathbf{W}_{q1}, \mathbf{W}_{q2}, \mathbf{W}_{qg}, \mathbf{W}_{k1}, \mathbf{W}_{k2}, \mathbf{W}_{kg}$ are the weights through dynamic tensor $\mathbf{W} \in R^{T \times S \times H \times H}$ decomposition that significantly reduce computational complexity. DCA incorporates dynamic tensor decomposition. This process reduces the number of parameters in the attention mechanism while preserving the model's ability to capture complex feature interactions. This is essential for real-time applications, where processing speed and accuracy are both critical.

2.4. Temporal convolutional network (TCN) and classification

Although the DCA module enhances the ability to dynamically adjust attention and highlight important temporal and spatial features, it does not explicitly model long-term temporal dependencies. The temporal dependencies in MI-EEG signals are deep and nonlinear, and traditional temporal convolutions have certain limitations in modeling long-term

dependencies in sequence data. Therefore, we use TCN, which consists of two residual block and dilated causal convolutions, capturing deeper temporal patterns as layers are stacked while avoiding potential bottlenecks in information flow and preserving multi-step temporal dependencies while expanding the receptive field. Therefore, the TCN module is introduced after the DCA to further strengthens the ability to capture long-range temporal dependencies within the sequence, which is crucial for understanding the temporal patterns in MI-EEG signals. The output of the TCN is the last element in the sequence. (More details on settings as described in [23].) The features are combined into a high-level feature representation through the Concatenate layer, and finally mapped to the specific classification tasks through the classification function (SoftMax). Then we adopt a compound loss strategy centered on categorical cross-entropy (CE) for multi-class MI-EEG classification. The CE loss is given by:

$$L_{CE} = - \sum_{a=1}^N y_a \log(\hat{y}_a) \quad (17)$$

where $y \in \{0, 1\}^N$ denotes one-hot encoded ground truth label, and $\hat{y} \in R^N$ represents the predicted class probabilities obtained through the softmax layer.

3. Experimental results and discussion

3.1. Datasets description and hyperparameters

The proposed model was evaluated using the BCI-2a dataset [25] and the PhysioNet EEG Motor/Imagery dataset [26]. In the BCI-2a dataset, a band-pass filter of 0.5–100 Hz and a notch filter at 50 Hz were applied to each channel. For the PhysioNet dataset, EEG signals were recorded at a sampling rate of 160 Hz. Due to annotation errors, data from participants 38, 88, 89, 92, 100, and 104 were excluded from the samples. Further details about these databases are provided in Table 1.

In this study, raw EEG signals were directly input into the model without additional preprocessing to preserve the authenticity of the data. To mitigate overfitting, improve robustness, and enhance the

Table 1

Description of all benchmark datasets.

Datasets	# Subjects	MI task	# C	Time points	MI trails
BCI-2a	9	Left hand, Right hand, Both feet, Tongue	22	1125	5184
PhysioNet	109	Left fist, Right fist, Both fists, Both feet	64	640	9241

generalization ability of the model, we also applied data augmentation techniques, such as time shifting, scaling, and adding noise to the raw data. Experimental results demonstrated the effectiveness of these methods.

The model was trained and tested using the TensorFlow framework on an Intel i9-12900K CPU and a single Nvidia GTX 3090 GPU (24 GB), with the Adam optimizer for training. Unless specified otherwise, all experiments in this paper utilized the hyperparameters shown in Table 2. These parameters were empirically set based on several experiments and remained fixed for all subjects.

3.2. Performance metrics and evaluation method

To assess the performance of our method for classifying MI-EEG data, we utilized measurement indicators including the accuracy, sensitivity (recall), specificity, precision, F1 and K-score (Cohen's kappa).

The accuracy metric is calculated as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (18)$$

where N is the total number of classes, TP and TN are true positive and true negative, and FP and FN are false positive and false negative, respectively.

The sensitivity shows the ability of our method to classify the positive trials correctly. We note that sensitivity and recall are mathematically identical metrics, both defined as:

$$Sensitivity(Recall) = \frac{TP}{TP + FN} \quad (19)$$

The specificity shows the ability of our method to classify the negative trials correctly.

$$Specificity = \frac{TN}{TN + FP} \quad (20)$$

The F1 is a reconciled average of precision and recall, which provides us with a combined metric of the precision and completeness of the

Table 2

Hyperparameter setting that used for all subjects.

Model	Parameters	Size	Parameters	Size
TSN	<i>Max_shift</i>	10	<i>Noise_level</i>	0.01
	<i>Scale</i>	0.1		
SCRC	Number of Channels (C)	22	Kernel size (K_{C2})	16
	Number of (E_2)	2	Number of Spatial filters	32
	Temporal filters (E_1)	7	(E_2)	1/8
	0.5		Split Ratio (α)	32
	Number of Channel filters (D)		Length of Time Vectors (d)	20
	Pooling size (P_2)		Number of Time Vectors (T_C)	
DCA	Gating Size (g)			
	Attention local heads	8	Number of windows (n)	2
	Attention global heads	4	Shared Channel Dimension (R)	5
Training Parameters	Learning Rate	0.0009	Epoch	1000
	Batch	64	Patience	300

model.

$$F1 = \frac{TP}{TN + (FP + FN)/2} \quad (21)$$

The precision refers to the quality of being exact or accurate. In the context of measurements, calculations, or predictions, precision indicates how closely the results agree with each other. It is a measure of the reproducibility or consistency of a set of values.

$$Precision = \frac{TP}{TP + FP} \quad (22)$$

The K-score is used for evaluation:

$$k_{score} = \frac{1}{N} \sum_{a=1}^N \frac{P_a - P_e}{1 - P_e} \quad (23)$$

where P_e is the observed agreement, P_a is the expected agreement, and N is the total number of classes.

For the BCI-2a dataset, we adopt both subject-dependent and subject-independent evaluation protocols. The subject-dependent setting strictly follows the official standard: training on session 1 (288 trials) and testing on session 2 (288 trials) for each subject. The subject-independent setting uses Leave-One-Subject-Out (LOSO) cross-validation to assess cross-subject generalizability.

For the PhysioNet dataset, we performed multi-subject validation using 10-fold cross-validation, considering the task heterogeneity and the inconsistent number of trials across subjects. In each fold, 90 % of all available trials were used for training and the remaining 10 % for testing. These protocols ensure comprehensive evaluation of model robustness across trials from multiple subjects.

3.3. Contributions of DCA-SCRCNet modules

This section evaluated the performance contributions of individual blocks within the DCA-SCRCNet model through ablation analysis. Table 3 presented the impact of removing one or more blocks from the DCA-SCRCNet model on MI classification performance using the BCI-2a dataset. The results indicated that both the DCA and SCRC modules contributed significantly to the performance, with overall accuracy improvements of 3.39 % and 1.29 %, respectively. When DCA and SCRC were excluded, the accuracy dropped. Specifically, adding only the DCA module resulted in a decrease of 0.76 % in accuracy, while adding only the SCRC module improved overall accuracy by 1.17 %. Including both DCA and SCRC increased accuracy by 1.72 %. Furthermore, the synergy between the SW and other modules was also analyzed. Without SW, adding only the DCA module caused a 0.73 % reduction in overall accuracy, while combining both TSN and the DCA module improved accuracy by 0.61 %. However, adding only the SCRC module decreased accuracy by 1.54 %, whereas adding both TSN and SCRC increased

Table 3

The contribution of each module in the DCA-SCRCNet model to motor imagery classification using the BCI-2a Dataset.

Removed block	Accuracy (%)	K-score
None (DCA-SCRCNet)	90.45	0.873
SCRC	89.16	0.856
TSN	88.75	0.850
DCA	87.06	0.827
TSN+SCRC	88.64	0.849
TSN+DCA	89.35	0.859
TSN+SCRC+DCA	88.73	0.849
SW	86.69	0.822
SW+SCRC	86.57	0.821
SW+DCA	86.11	0.815
SW+TSN +SCRC	85.96	0.813
SW+TSN +DCA	85.15	0.802
SW+TSN +SCRC+DCA	88.37	0.844

accuracy by 0.96 %. This suggested that SW, TSN, SCRC, and DCA work in tandem to enhance both the accuracy and generalization ability of the model. Removing SW, TSN, and the SCRC module led to a significant 4.49 % drop in accuracy, while removing SW, TSN, and the DCA module resulted in an even larger drop of 5.3 %. These findings indicate that each block contributes to the classification performance, especially the collaborative effects of SW, TSN, SCRC, and DCA, which are critical components in the success of the DCA-SCRCNet model.

3.4. Parameter analysis

In this section, we investigated the impact of variations in the spatial and channel reconstruction convolution in the SCRC module, particularly focusing on the channel division ratio of the feature map F^o , on the performance of the DCA-SCRCNet model using the BCI-2a and PhysioNet datasets. The feature map F^o was compressed using convolution to reduce the number of channels, which were then ranked by weight size and divided according to the set ratio. Channels were classified into αC and $(1 - \alpha)C$, where α represented the division ratio.

As shown in Fig. 4, we evaluate the classification performance under various channel partition fractions. Here, the ratio (e.g., 1/8) denotes the fraction of the total channels retained as important, rather than a strict numerical ratio between important and redundant channels. The overall best performance is achieved when retaining

1/8 of the channels, which effectively balances spatial redundancy removal and information preservation. On the PhysioNet dataset, classification accuracies under 1/8, 2/8, and 3/8 are nearly identical, suggesting a performance plateau when retaining up to 3/8 of the channels.

Despite this, 1/8 is still preferable due to its lower complexity and equivalent or better accuracy, thus selected as the optimal setting. In contrast, on the BCI-2a dataset, the performance sharply peaks at 1/8, with minor rebounds at 3/8 and 6/8. This inconsistency may stem from dataset-specific spatial feature distributions and does not generalize across datasets.

We also explored an approach to further enhance DCA performance: adjusting the local-to-global attention head ratio, as proposed in [27]. We empirically set a fixed number of global heads and feature dimensions, and varied the ratio of local to global heads. Despite differences in the number of local and global heads, the model dynamically adjusted its attention distribution based on the specific EEG signals, thereby finding the optimal balance between local and global information.

As shown in Table 4, different ratios significantly impacted the classification performance of the model. Through extensive experimentation with various combinations of local and global head counts, we found that a local-to-global attention head ratio of 2:1 yielded the best classification results, as shown in Fig. 5. The dynamic compositional mechanism of DCA allowed the model to adjust the combination of attention heads under different input conditions. This flexibility meant

Table 4

The impact of different head ratio on model performance.

Local: Global Attn.	1:1	2:1	3:1	4:1	5:1	7:1
Accuracy (%)	89.68	90.45	89.51	89.35	88.04	71.55
K-score	0.86	0.87	0.86	0.86	0.84	0.62

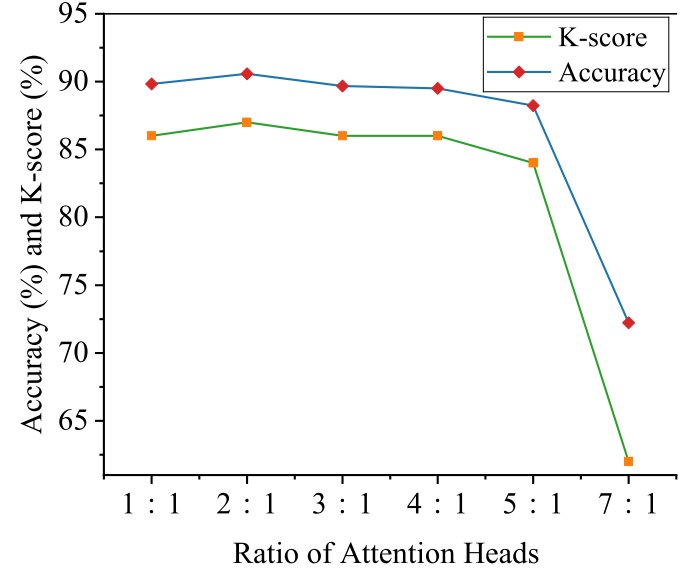


Fig. 5. Comparison chart of different attention heads ratio configurations.

that whether the number of global heads or feature dimensions remains constant, adjusting the ratio of local to global heads affects how the model balances the capture of local details and global structures. This was particularly important in complex MI-EEG tasks, where it directly influenced the decoding accuracy and computational complexity.

Fig. 6 showed the accuracy distribution for different subjects (S01-S09). Each violin plot illustrates not only the median accuracy (dashed line) but also the range of accuracy distributions (25 %-75 % inter-quartile range). The accuracy differences highlighted subject-specific variations in MI-EEG signals. DCA adjusted attention weights dynamically, adapting to individual local features, reducing decoding performance fluctuations. DCA captured local features specific to each subject,

improving individualized decoding performance. For example, higher accuracy in S03 and S08 might be due to effective capture of their stable signal characteristics in DCA. Dynamic projection reduced noise and compresses redundant features, resulting in narrower accuracy distributions for some subjects (e.g., S07, S08), indicating better feature extraction.

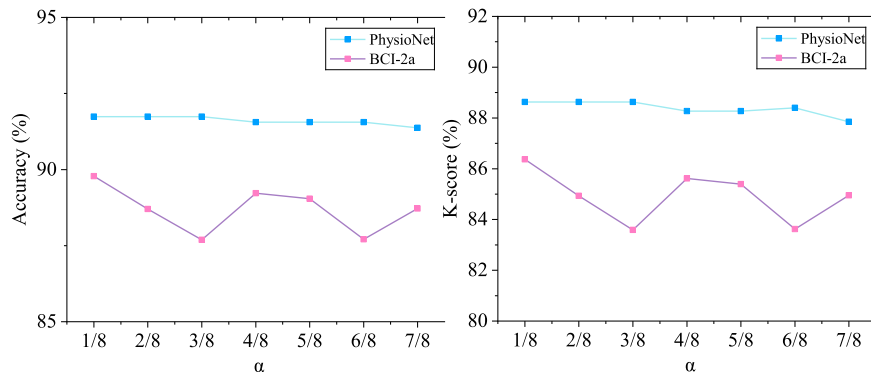


Fig. 4. Impact of channel partition ratio α on model performance for both datasets.

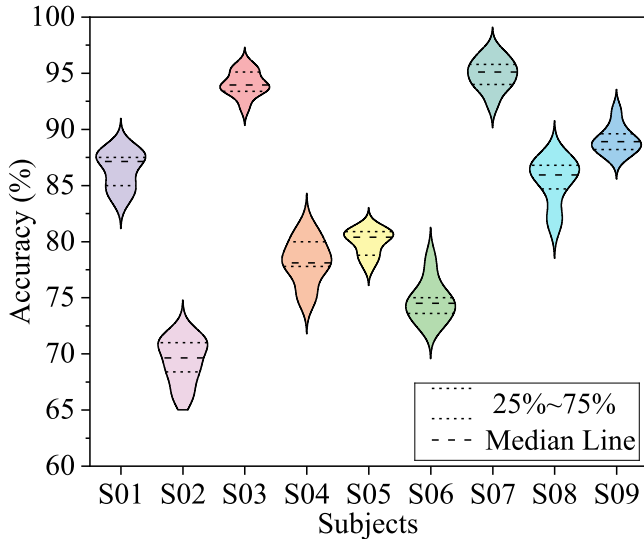


Fig. 6. Distribution of classification accuracy across multiple trials for 9 subjects.

3.5. Comparison to different attention schemes and recent studies

In Table 5, we compared the performance of the proposed model with five different attention mechanisms: MSA [28], SIMAM [29], SE [30], CBAM [31], and MHLA [32]. For MSA, the number of heads was set to 2, with each head size set to 8. The reduction ratio for both SE and CBAM was experimentally set to 8. The results showed that all attention mechanisms improve the performance of the DCA-SCRCNet model, with the DCA model yielding the best performance, indicating that the DCA model is particularly suited for representing 2D EEG signals. In this section, we evaluate the performance of the DCA-SCRCNet model using the BCI-2a dataset, comparing it with other baseline models, including EEGNet [11], MBEEG_SENet [33], EEG-TCNet [21], TCNet_Fusion [34], ATCNet [23], and DB-ATCNet [35]. Preprocessing, training, and evaluation procedures were performed according to the methodology specified in this paper, with the results based on the parameters defined in the original papers. Among them, the DCA module achieved the highest overall accuracy (90.45 %) and K-score (0.8727), indicating superior classification and

agreement performance across subjects. Importantly, DCA also achieved the best F1 (0.89), demonstrating strong balance between precision and recall, and suggesting robustness in identifying both positive and negative classes in imbalanced or noisy data. The recall of 0.89 further indicates DCA's strong ability to correctly identify true motor imagery events, which is critical in BCI applications.

Although SE and SIMAM also showed strong performance (e.g., SE F1 = 0.88), their improvements were slightly inferior to DCA. Notably, SIMAM uses fewer parameters (105.9k) but yielded lower recall (0.78) and F1 (0.78), suggesting a trade-off between compactness and discriminability. MSA and MHLA introduced

additional parameters (~116.2k) but still underperformed DCA in

most metrics. Table 6 presented the performance of each model across 10 random runs. The results showed that DCA-SCRCNet achieves an average accuracy of 90.45 % and a K-score value of 0.87, outperforming all other models across all subjects. This demonstrated that the model not only excels in learning ability but also offered superior stability in performance across multiple runs. This box-plot and mean line chart in Fig. 7 compares the performance of different models on the BCI-2a dataset. Each colored box represents a model, and the orange one is the proposed model. The vertical axis shows the values of performance indicators. The median line inside each box divides

the data in half, and a higher median of the orange box indicates better overall performance of the proposed model. The height of the box, the inter-quartile range (IQR), reflects the middle 50 % of the data; a smaller IQR for the orange box means more consistent performance. The whiskers represent the spread of data outside the IQR, with longer whiskers indicating greater variability. Outliers, shown as points outside the whiskers, may suggest extreme performance cases of the model. The connecting lines represent the average values of the indicators. Comparing the position of the orange mean line with others reveals the average performance advantage or disadvantage of the proposed model.

As shown in Table 7, our DCA-SCRCNet model, without pre-processing, demonstrates excellent performance in both subject-dependent and subject-independent evaluations, especially in subject-independent evaluations, where it showcases its strong generalization ability to new subjects. In conclusion, DCA-SCRCNet's outstanding performance on the BCI-2a dataset validates its potential and advantages as an MI-EEG classification model. Through comparative experiments with several SOTA models, DCA-SCRCNet outperforms existing models in classification performance, learning ability, and stability, demonstrating its exceptional performance in MI decoding tasks.

Additionally, we tested the generalization capability of the DCA-SCRCNet model on the PhysioNet MI-EEG dataset for two-class (left fist and right fist) and four-class (left fist, right fist, both fists, and both feet) classification tasks under a multi-subject validation protocol. As shown in Table 8, we compared the performance of DCA-SCRCNet with other SOTA models on the same dataset. The proposed model achieved accuracies of 92.7 % and 76.2 % for the two-class and four-class classification tasks, respectively. Compared to existing models, the proposed model improved accuracy by 2.6 % in the two-class task and 5.8 % in the four-class task.

Table 9 presents the subject-specific classification performance of the proposed model on the PhysioNet MI-EEG 2-class dataset, evaluated across 103 subjects. The model achieves an average accuracy of 83.2 %, with a maximum accuracy of 92.7 % on individual subjects. The average sensitivity and specificity are both 0.82, indicating balanced detection of positive and negative classes. The F1-score, which

reflects overall classification robustness, reaches 0.83 on average. These results demonstrate the model's ability to generalize well across individuals, despite physiological variability, and confirm its suitability for real-world subject-specific MI-EEG decoding.

To quantitatively evaluate the trade-off between performance and model complexity, we compare our proposed DCA-SCRCNet against several representative state-of-the-art architectures in terms of parameter count, classification accuracy, and K-score, as summarized in Table 10. Our model achieves the highest subject-specific accuracy

Table 5

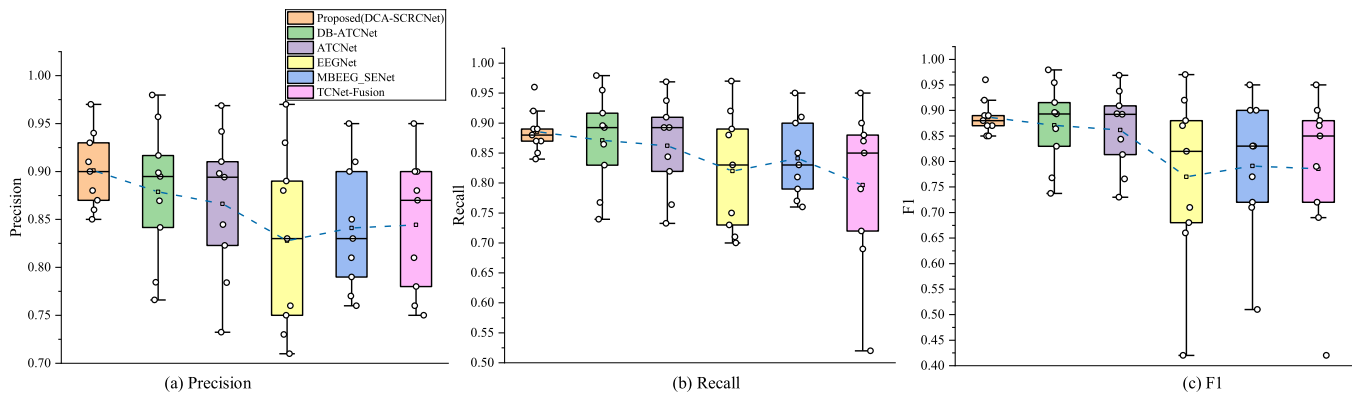
The DCA-SCRCNet model exhibits varying performance with different attention mechanisms.

Attention mechanism	Accuracy (%)	K-score	Precision	Recall	F1	Parameters
No Attention	87.06	0.8274	0.87	0.87	0.87	105.3k
MSA [28]	88.77	0.8503	0.88	0.86	0.85	116.2k
SIMAM [29]	89.64	0.8619	0.79	0.78	0.78	105.9k
MHLA [32]	88.55	0.8470	0.89	0.86	0.86	116.2k
CBAM [31]	88.70	0.8493	0.89	0.78	0.87	106.2k
SE [30]	89.56	0.8609	0.89	0.88	0.88	105.7k
DCA	90.45	0.8727	0.90	0.89	0.89	117.6k

Table 6

Comparison of performance (Accuracy (%) and K-score) for subject-specific classification using the BCI-2a dataset: the proposed model vs. other reproduced models.

	DCA-SCRCNet (Ours)		DB-ATCNet [35]		ATCNet [23]		EEGNet [20]		MBEEG_SENet [33]		TCNet-Fusion [34]	
Subject	Accuracy (%)	K-score	Accuracy (%)	K-score	Accuracy (%)	K-score	Accuracy (%)	K-score	Accuracy (%)	K-score	Accuracy (%)	K-score
1	92.0	0.89	91.3	0.88	92.0	0.89	89.6	0.86	89.6	0.86	87.2	0.83
2	83.0	0.77	73.6	0.66	81.4	0.75	66.3	0.55	78.3	0.71	66.0	0.55
3	97.1	0.96	97.2	0.97	97.7	0.97	96.9	0.96	94.8	0.93	93.8	0.92
4	86.8	0.82	85.1	0.81	76.7	0.69	70.8	0.61	83.2	0.78	74.3	0.66
5	88.4	0.84	83.5	0.78	85.4	0.81	79.9	0.73	86.0	0.81	79.9	0.73
6	86.8	0.82	79.0	0.72	85.4	0.81	65.3	0.54	75.2	0.67	66.3	0.55
7	94.4	0.93	95.2	0.94	68.1	0.57	90.3	0.87	56.3	0.42	92.7	0.91
8	92.4	0.90	90.1	0.88	93.6	0.91	86.8	0.82	90.6	0.88	88.5	0.85
9	93.2	0.91	91.4	0.86	91.2	0.88	88.2	0.84	90.3	0.87	88.5	0.85
Mean	90.5	0.87	87.2	0.83	85.7	0.81	81.6	0.75	82.7	0.77	81.9	0.76

**Fig. 7.** Comparison of model performance indicators on BCI-2a dataset: box-plot and mean line chart.**Table 7**

Comparison of DCA-SCRCNet and state-of-the-art methods on the BCI-2a dataset.

Study	Description Performance					
	Input data ^a	DL approach	Subject-dependent		Subject-independent	
			Accuracy (%)	K-score	Accuracy (%)	K-score
Lawhern et al. [20]	RS	CNN (EEGNet)	80.1	0.73	68.8	0.58
Hassanpour et al. [8]	SF	DBN-AE	71.0	-	-	-
Amin et al. [9]	RS	Multi-layer-CNN and AE	75.0	-	55.3	-
Ingolfsson et al. [16]	RS	CNN+TCN (EEG-TCNet)	80.0	0.73	69.5	0.59
Musallam et al. [34]	RS	Multi-layer CNN, TCN (TCNet Fusion)	80.9	0.75	70.6	0.61
Amin et al. [21]	RS	Attention, inception CNN, LSTM	82.8	-	-	-
Altuwaijri et al. [34]	RS	Attention, multi-branch CNN	82.2	0.76	68.7	0.58
Zhi et al. [37]	RS	Multi-domain CNN (TSFCNet)	82.7	0.77	-	-
Altaheri et al. [23]	RS	Attention, CNN, TCN (ATCNet)	85.7	0.81	70.9	0.61
Qin et al. [36]	RS	CNN, Multi-Feature Attention (M-FANet)	79.3	0.73	-	-
Zhou et al. [35]	RS	Dual-branch, CNN, Attention, TCN (DB-ATCNet)	87.5	0.83	71.3	0.62
Yang et al. [38]	RS	Multi-branch CNN, Attention, TCN (MCMNet)	86.4	0.82	-	-
Han et al. [39]	RS	Multi-scale CNN, Attention, TCN (MSAFNet)	88.1	0.84	68.8	0.58
Proposed method	RS	SCRC, DCA, CNN, TCN (DCA-SCRCNet)	90.5	0.87	70.7	0.61

^a Input formulation: Spectral features (SF), Raw signal (RS).

(90.5 %) and K-score (0.87), outperforming all baseline methods.

In terms of model size, DCA-SCRCNet (117.6k parameters) surpasses larger models such as DB-ATCNet (149.7k, 87.2 %) and Deep ConvNet (283.3k, 82.1 %), demonstrating that our spatial-channel reconstruction and dynamic attention mechanisms yield better accuracy without incurring excessive computational overhead. Compared to ultra-lightweight models like EEGNet (2.6k, 81.6 %), our model maintains competitive compactness while avoiding performance degradation, confirming that overly aggressive compression may compromise discriminative capacity.

Importantly, inference time—a critical factor in real-time motor imagery (MI) applications—was also considered. DCA-SCRCNet

achieves a runtime of 0.7117 s on standard hardware, falling within the real-time latency range (typically under 1–2 s) considered acceptable in practical MI-BCI systems [50]. Despite maintaining real-time responsiveness, we still leaves room for further improvement compared with the 0.045 s latency demonstrated in the hand-open/close MI-BCI of Giri et al. [51]. These results validate that DCA-SCRCNet provides an effective balance between accuracy, model compactness, and real-time inference capability, which is essential for reliable deployment in real-world MI-EEG decoding tasks.

DCA-SCRCNet demonstrated superior classification performance across all motor imagery tasks. Fig. 8(a) and (b) presents confusion matrices for different models, showing that DCA-SCRCNet improved

Table 8

Comparison of DCA-SCRCNet and state-of-the-art methods on the PhysioNet dataset.

Models	Subjects	Class Type ²	Max. Accuracy ³	Avg. Accuracy ⁴
SUT-CCSP+SVM (2013) [40]	56	MI 2-class	90.0 %	72.4 %
Phase information (2014) [41]	103	MI 2-class	71.6 %	-
SUT-CCSPRandom forest (2016) [42]	24	MI 2-class	-	80.1 %
IMOCs-85 subjects (2016) [43]	85	MI 2-class	-	62.8 %
IMOCs-35 subjects (2016) [43]	35	MI 2-class	-	79.9 %
CNNs (2018) [44]	105	MI 2-class	-	80.4 %
G-CRAM (2020) [45]	105	MI 2-class	74.7 %	-
EEGNet Fusion (2020) [46]	103	MI 2-class	-	83.8 %
BENDR (2021) [47]	105	MI 2-class	-	86.7 %
S-CTrans (2022) [48]	109	MI 2-class	-	83.3 %
DB-ATCNet (2024) [35]	103	MI 2-class	90.1 %	79.5 %
CNNs (2018) [44]	105	MI 4-class	-	58.6 %
EEGNet(2020) [20]	105	MI 4-class	-	65.1 %
ConTraNet CNN-Transformer (2022) [49]	105	MI 4-class	-	65.4 %
T-CTrans (2022) [48]	109	MI 4-class	-	68.5 %
DB-ATCNet (2024) [35]	103	MI 4-class	70.4 %	69.6 %
Proposed method-2 class	103	MI 2-class	92.7 %	83.2 %
Proposed method-4 class	103	MI 4-class	76.2 %	62.0 %

²Class type: number of MI classes used. ³Max. Accuracy: maximum classification accuracy. ⁴Avg. Accuracy: average classification accuracy. In our study, Max. Accuracy and Avg. Accuracy are the best accuracy and average accuracy of 10 experiments, respectively.

Table 9

Subject-Specific Performance Comparison on PhysioNet MI-EEG (2-Class Task).

Proposed method	Subjects	Class Type	Max	Avg
Accuracy	103	MI 2-class	0.93	0.83
Sensitivity	103	MI 2-class	0.91	0.82
Specificity	103	MI 2-class	0.91	0.82
F1	103	MI 2-class	0.92	0.83

Table 10

Comparative analysis of state-of-the-art algorithms for subject-dependent classification.

Network Architecture	Parameters	Inference time	Accuracy (%)	K-score
Shallow ConvNet [10]	46.1k	0.6996 s	79.7	0.73
Deep ConvNet [10]	283.3k	0.6482 s	82.1	0.75
EEGNet [20]	2.6k	0.4252 s	81.6	0.75
TCNet-Fusion [34]	17.6k	0.5779 s	81.9	0.76
MBEEG_SENet [33]	10.5k	0.7445 s	82.7	0.77
ATCNet [23]	115.2k	0.4548 s	85.7	0.81
DB-ATCNet [35]	149.7k	0.6325 s	87.2	0.83
DCA-SCRCNet (Ours)	117.6k	0.7117 s	90.5	0.87

classification performance across all MI classes. The results indicated that the DCA-SCRCNet model performs well in recognizing left fist, right fist, feet, and both fist categories. This suggests that the model effectively captured neural activity in the bilateral brain regions during unimanual motor tasks, addressing the current gap in recognizing neural activity

during bimanual movements. Additionally, the model excelled in processing neural activity from the motor cortex on the ipsilateral side. In the two-class classification tasks, the extracted features showed high separability, which indicated that DCA-SCRCNet can efficiently capture the features of these two motor imagery tasks. However, in the four-class classification task, the features extracted from the feet movement imagery exhibit higher separability, while those from both fists overlap significantly, showing lower separability, which impedes accurate classification.

In the two-class classification tasks, the extracted features show high separability, which indicated that DCA-SCRCNet can efficiently capture the features of these two motor imagery tasks. However, in the four-class classification task, the features extracted from the feet movement imagery exhibit higher separability, while those from both fists overlap significantly, showing lower separability, which impedes accurate classification. In summary, outstanding performance of DCA-SCRCNet on the small-sample BCI-2a dataset and the large-sample PhysioNet dataset validated its potential and advantages as an MI-EEG classification model. Through comparative experiments with various SOTA models, DCA-SCRCNet outperforms existing models in classification performance, learning ability, and stability, demonstrating its exceptional performance in MI decoding task.

3.6. Visualization

In this section, we visualized the evaluation results of the DCA-SCRCNet model based on the BCI-2a dataset and the PhysioNet MI-EEG dataset. T-SNE (t-distributed stochastic neighbor embedding) [52] is a widely used statistical technique for dimensionality reduction and feature visualization. It mapped high-dimensional data points to a lower-dimensional space while preserving the local structure of the data. Fig. 9 illustrates the feature distribution of the reproduced models and DCA-SCRCNet after training on the BCI-2a dataset. The visualization results showed that the features extracted by DCA-SCRCNet have clearer boundaries and more distinct clustering, further demonstrating the model's advantage in effectively extracting feature information and enhancing class discrimination.

Fig. 10 illustrates the spatial feature distributions before and after processing by the proposed model, using data from five subjects in the BCI Competition IV-2a dataset. The raw EEG maps (top row) show diffuse and low-contrast patterns, indicating spatial redundancy and weak task-related activations due to noise and volume conduction.

In contrast, the processed feature maps (bottom row) highlight more localized and structured activations, especially in sensorimotor regions (e.g., C3/C4), which are relevant for motor imagery tasks. These maps exhibit enhanced contrast and inter-subject consistency, suggesting that the model effectively filters irrelevant spatial noise while enhancing task-discriminative patterns. This demonstrates the model's ability to extract subject-adaptive and neurophysiologically meaningful features, contributing to improved decoding performance.

In the subject-dependent task, we selected two subjects (S5 and S9) at random for visualization of the training and validation accuracy and loss curves. During the experiment, we employed an early stopping strategy to automatically halt training when the accuracy and loss curves converged within the maximum epoch, ensuring the reliability of the experiment while saving computational resources. As shown in Fig. 11, the validation losses for the three subjects did not show signs of overfitting, and the model was able to adjust the epoch size based on individual subject performance, saving computational resources. This indicated that our model is highly effective for MI classification.

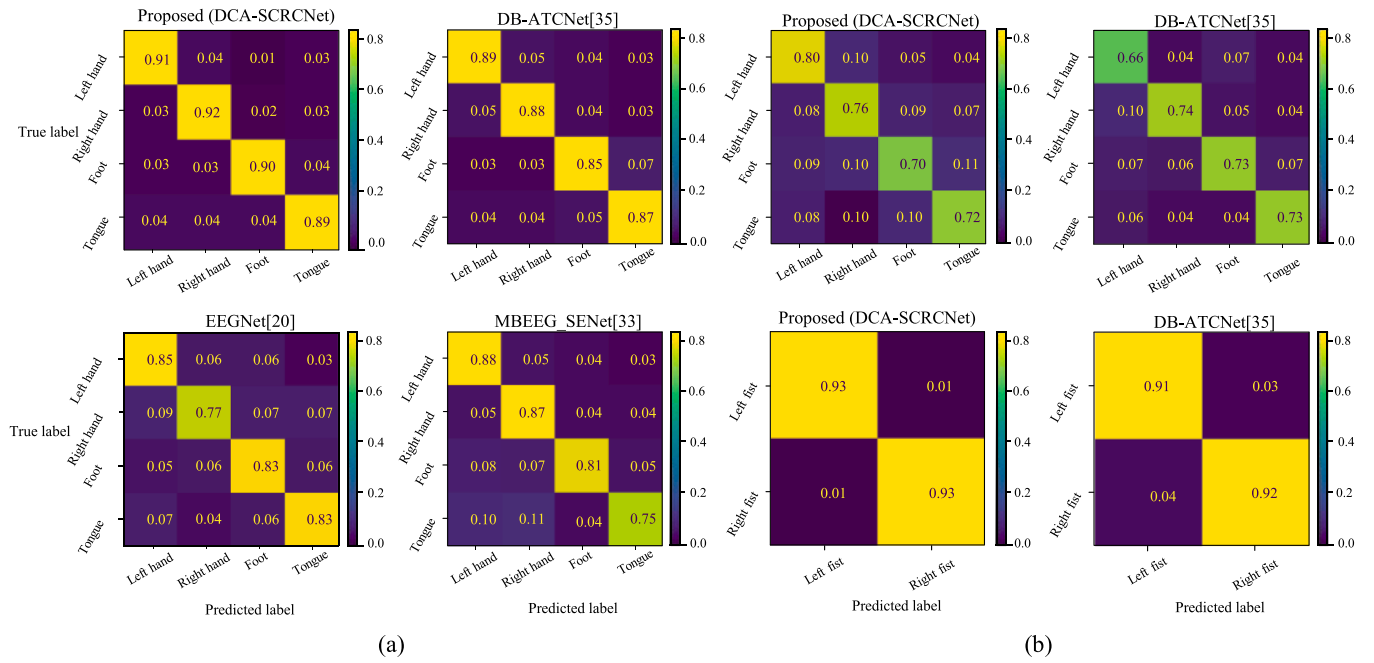


Fig. 8. (a) Confusion matrices of the reproduced models and DCA-SCRCNet in BCI-2a dataset. (b) Confusion matrix of DCA-SCRCNet and DB-ATCNet for two-class and four-class classification tasks on the PhysioNet dataset.

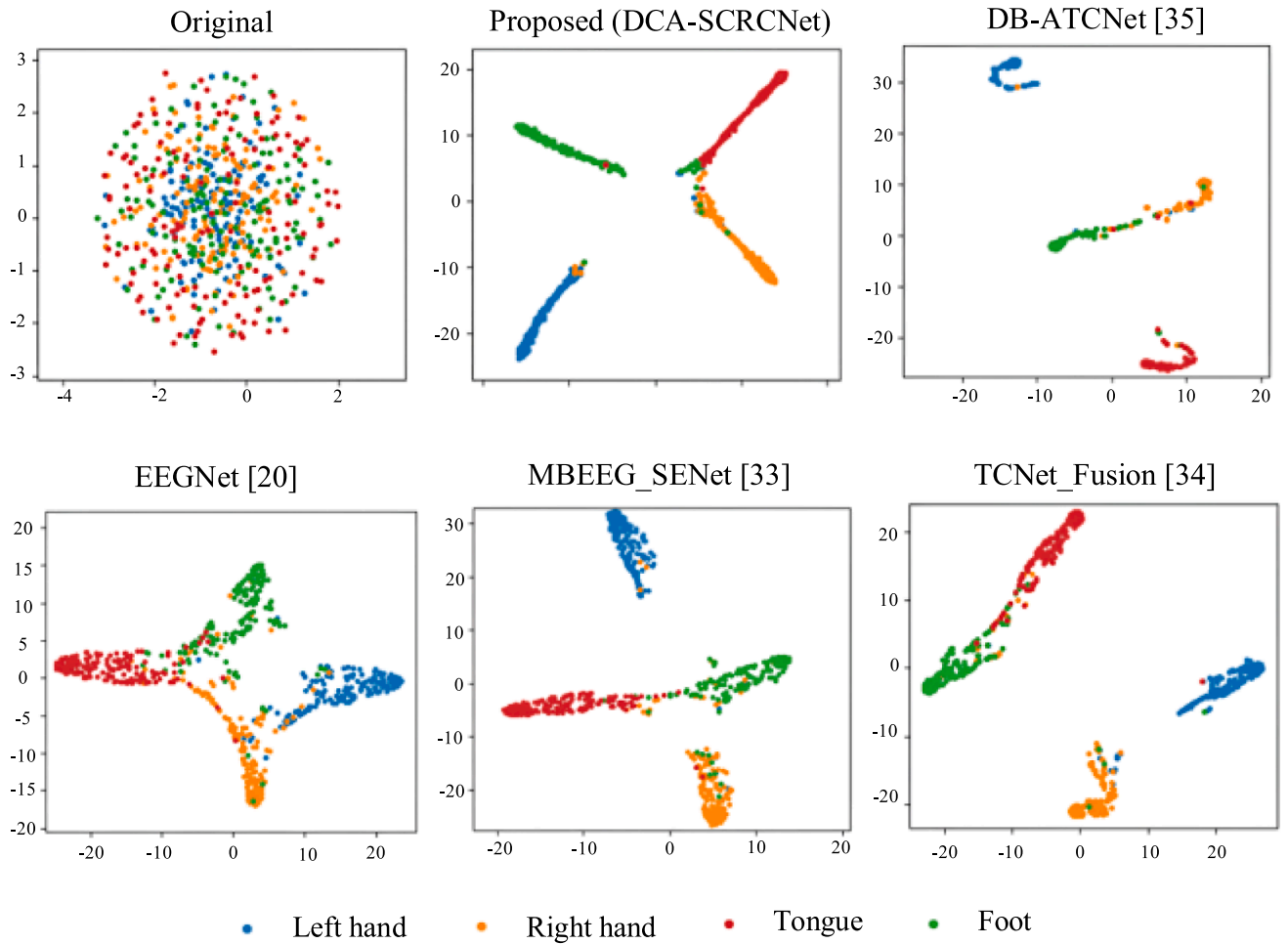


Fig. 9. The results of t-SNE visualization of the extracted features using the reproduced networks and DCA-SCRCNet for the subject-specific classification task on the BCI-2a dataset.

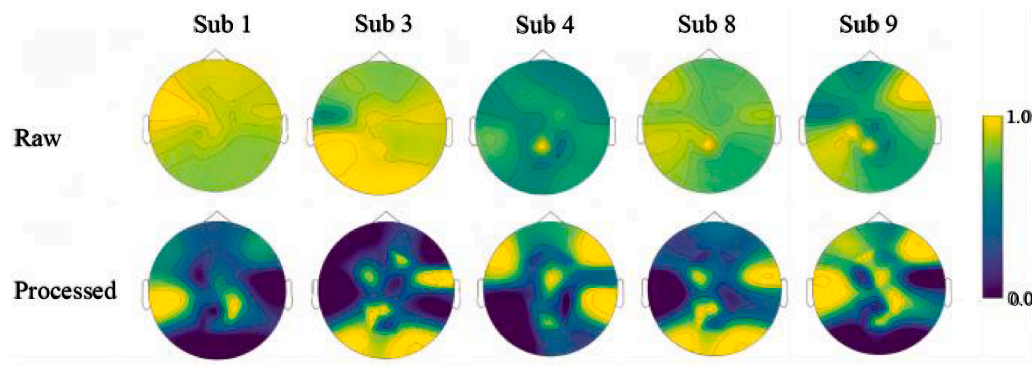


Fig. 10. Spatial feature maps before and after model processing on BCI-IV 2a dataset.

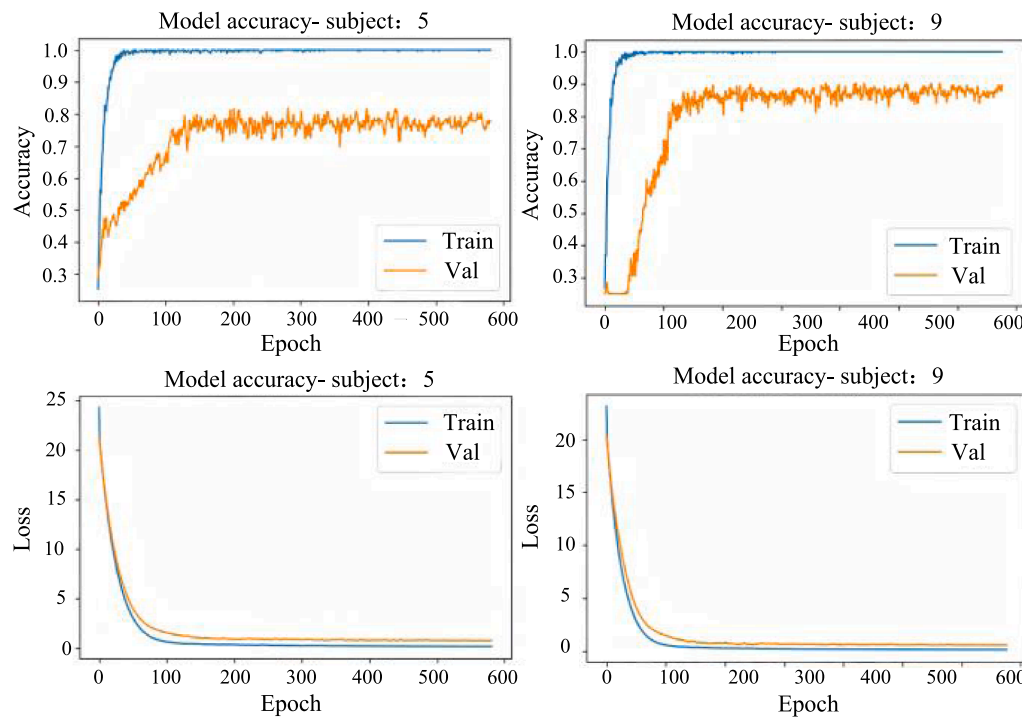


Fig. 11. Performance accuracy and training and validation losses for Subjects S5 and S9 during the training period of BCI-2a dataset for subject-dependent four classification.

4. Conclusion

This paper presents a dynamic compositional attention-based spatial reconstruction convolutional network (DCA-SCRCNet) model for recognizing MI activities from EEG signals. Ablation experiments demonstrate the contribution of each module to the overall performance of the DCA-SCRCNet model. When evaluated on the challenging BCI-2a dataset, with minimal preprocessing and without artifact removal, the model achieved a subject-independent accuracy of 71.34 % and a subject-dependent accuracy of 90.5 %, outperforming current DL architectures. This highlights the powerful ability of DCA-SCRCNet to recognize MI activities directly from raw EEG signals. The DCA-SCRCNet model significantly enhances EEG decoding performance for all subjects in the BCI-2a dataset, demonstrating its ability to extract generalized MI features from different categories and subjects. Furthermore, on the PhysioNet dataset, the model achieved strong performance in both two-class and four-class classification tasks, further validating its impressive trial-level robustness and feature recognition capabilities. DCA-SCRCNet combines high performance with relatively fewer parameters, making it well-suited for resource-constrained Internet of Things

(IoT) and edge devices. Despite the promising results, there are some limitations that hinder its practical application in real-world settings. One limitation is the length of the time window, which is currently too long. Future work will focus on reducing the time window length used by DCA-SCRCNet and other CNN-based BCI methods (4.5 s for the BCI-2a dataset and 4 s for the PhysioNet dataset), as these long windows introduce latency in online BCI applications. Additionally, we will address errors arising from large differences in subject samples to further enhance the model's generalization ability. Despite these challenges, the continuation of this work will contribute to the future development of affordable portable EEG-based BCI systems.

CRediT authorship contribution statement

Jiayin Chen: Validation, Supervision, Formal analysis. **Xiaoyu Zou:** Writing – review & editing, Validation, Investigation. **Jiajing Song:** Validation, Investigation. **Liwen Zhang:** Software, Investigation. **Zhijie Dong:** Writing – original draft, Software, Methodology. **Pengjun Zhao:** Writing – review & editing, Supervision, Project administration. **Jie Pan:** Writing – review & editing, Writing – original draft, Methodology,

Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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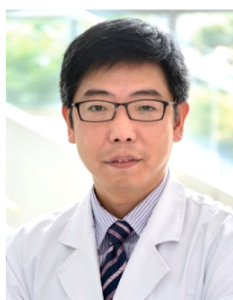
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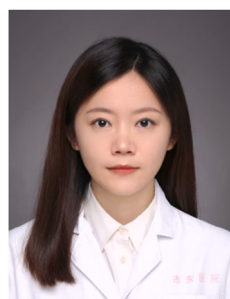


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